Phase 3 Project Submission: Core Implementation & Validation

1. Implementation of Core Functionalities

- **Completed Core Modules:** Full implementation of the primary research objective (e.g., custom cryptographic algorithms, ZK-SNARK circuits, core smart contracts, or P2P network protocols).
- **Codebase Submission:** A clean, heavily commented codebase containing the integrated core functionalities. The code must be runnable.

2. System Integration and Testing Environment

- **Component Integration:** Successful integration of off-chain logic (e.g., Provers, backend servers, indexing scripts) with on-chain components (e.g., EVM smart contracts, local testnet).
- **Execution Logs & Test Suites:** Evidence of initial testing and debugging. You must provide terminal logs, transaction hashes (on testnet), or output files proving the components function together and comply with Phase 2 requirements.

3. Performance Metrics and Benchmarking (Crucial for Research)

- **Data Collection:** Collection of preliminary, hard performance data. Depending on your project, this must include specific blockchain/cryptographic metrics (e.g., Gas consumption, computational overhead, proof generation/verification time, Transactions Per Second (TPS), or memory usage).
- **Comparative Validation:** Validation of your system against predefined benchmarks. If you are proposing a new solution, compare its performance metrics directly against the existing standard (e.g., comparing your SHA-1024 implementation's CPU time against standard SHA-512).

4. Documentation and Reproducibility

- **Updated Architecture:** Finalized system architecture diagrams or flowcharts reflecting the actual implemented design.
- **Reproducibility Instructions:** A detailed `README.md` containing strict, step-by-step instructions on how to set up the environment, install dependencies, and run the code. (If the Professor or TA cannot run your project using these instructions, it will be considered incomplete).

5. IEEE Research Paper: Initial Draft & Insights

- **Preliminary Report:** A summary of key achievements in Phase 3 and the engineering roadblocks you resolved.

- **IEEE Draft (Methodology & Setup):** A highly detailed draft of the **"Proposed Methodology"** and **"Experimental Design"** sections of your final IEEE research paper. This must clearly explain the mathematical/architectural setup of your experiments so they are ready for the final Phase 4 report.

6. Security Analysis & Edge Cases

- **Threat Modeling:** Identification of potential vulnerabilities in your current implementation (e.g., reentrancy risks in contracts, collision vulnerabilities in hashes, centralization bottlenecks).
- **Edge Case Testing:** Analysis of how your system handles extreme or unexpected inputs (e.g., network latency, malformed data, or boundary conditions in mathematical proofs).

7. Plans for Phase 4 (Final Polish & Presentation)

- **Completion Roadmap:** A detailed outline of the remaining tasks required to finalize the IEEE research paper, polish the codebase, and prepare for the final project presentation.